



What is the technology that makes BLDC fans expensive?

When comparing a regular AC induction motor to a brushless DC motor, some notable distinctions arise. The AC induction motor relies on copper windings to generate the necessary magnetic fields for rotation. In contrast, a brushless DC motor features copper windings, magnets, and crucially, electronic hardware equipped with a microcontroller. It's this additional electronic hardware and microcontroller that represent the primary cost differential. In terms of fundamental components, apart from this added hardware, the two motor types are quite comparable.

Furthermore, the current market demand continues to favour traditional fans over their more efficient counterparts. Consequently, the production volumes of energy-efficient fans have remained relatively low.

Is India a net creator or a net consumer for fans?

The shifting dynamics of the Indian fan market and the increase in imports of BLDC fans have sparked a debate. Until recently, India dominated its fan market, accounting for about 99% of fan production.

This shift raises questions and concerns, which we are actively working to address. India has a wealth of experience in fan manufacturing, producing over forty million fans annually, a significant number even on a global scale. The innovations brought about by Indian companies, like our own, have the potential to reshape the global fan industry. This stands in contrast to other industries, such as air-conditioners, where foreign brands have pushed for energy efficiency standards in India to promote their own products and technology.

In the case of fans, India has both the core technology and

manufacturing capabilities. While some electronic components may still be imported, the fundamental technology and expertise reside in India. Leveraging this capability effectively could position India as a global producer of fans, and our company is uniquely positioned in this regard. Unlike other brands that rely on outsourcing components, our company possesses the core technology, handles manufacturing, and owns the brand, allowing for direct engagement with the market to meet consumer needs.

The global demand for fans is universal, making India's potential as a major fan producer an exciting prospect. This aligns with our vision of utilising India's manufacturing prowess and technological innovation to make a significant impact on the global fan market.

Can you point out and define the key elements that need to be worked upon for India to become a net creator for fans?

Meeting the global demand for fans through rapid manufacturing scale-up is a critical requirement. Achieving this scale swiftly, especially in a competitive market alongside countries like China and Southeast Asian nations, poses a distinct challenge. While India possesses the core technology and fan production knowledge, achieving the necessary scale at an accelerated pace is no straightforward feat.

Manufacturing systems that can efficiently produce millions of super energy efficient fans must be established. Additionally, the ability to adapt to changing consumer preferences and aesthetics is essential, particularly in Western markets where fan usage is different and aesthetics play a significant role alongside functionality.

The challenge lies in bridging the gap between having the technology

and knowledge and rapidly scaling up production to meet global demand. This involves not only expanding manufacturing capacity but also ensuring flexibility to accommodate changing consumer preferences and market trends.

Is there any difference between the BLDC technology made in India and the ones that are manufactured and developed in other countries?

While there may be various product-level differences, the core technology behind BLDC fans remains consistent, with variations tailored to meet specific market demands. The primary difference between fan requirements in India and Western countries lies in their intended use. In India, most fans are designed to operate at higher speeds, typically around 350-400 RPM, to deliver optimal comfort for the local demographic.

Regarding BLDC (brushless direct current) fans, the fundamental technology remains the same. These fans consist of a magnet-based motor, coils, and electronics. However, there can be variations in how this technology is implemented, resulting in different product offerings. The distinctions primarily manifest at the product level, including factors such as blade shape and length, rather than significant differences in the internal technology.

What does the overall demand and supply for BLDC fans in the past two or three years look like, and the key factors that have pushed it forward?

One of the primary driving forces for adopting more energy-efficient fans in India has been government mandates, particularly the introduction of star labelling regulations. This initiative, which came into effect on January 1, 2023, is a significant push towards energy